

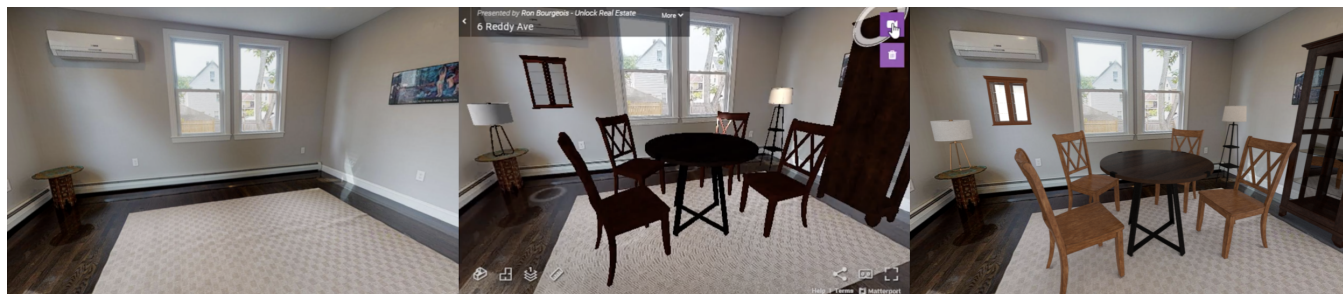


# Home Studio: DIY Interior Design in Mixed Reality

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**Figure 1: Home Studio enables a drag-and-drop furnishing experience, letting customers create photo-realistic renders of Wayfair products at the property scale and jump in for an immersive virtual reality experience.**

## ABSTRACT

Virtual staging of real estate listings increases the appeal of a property by letting prospective buyers envision a living space remotely. However, existing tools employed to stage homes limit the scale of the visualization to a set of fixed images provided by customers or require 3D artist expertise to reconstruct the space. The adoption of 3D Matterport scans has accelerated due to the Covid-19 pandemic as a means to enable virtual tours and adhere to social distancing guidelines. We present Home Studio, a virtual staging tool that empowers non-experts, letting them furnish any Matterport scene and create photo-realistic renders in a matter of minutes. Our tool lets customers dive into their designs using a virtual reality headset to assess the final product in an immersive experience.

## CCS CONCEPTS

• **Applied computing** → *Online shopping*; • **Computing methodologies** → *Mixed / augmented reality*; Rendering.

## KEYWORDS

Mixed Reality; Rendering; Interior Design; E-Commerce

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## 1 INTRODUCTION

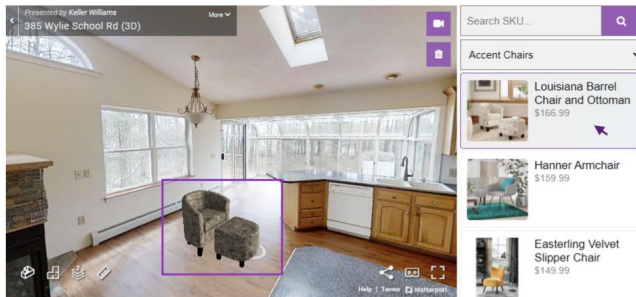
Contextualization of purchases in spaces is crucial to bridging the customer imagination gap, increasing trust that products will fit in both style and substance in their own homes [Space10 2017]. The impact of product visualization is especially apparent in real estate, with staged homes selling 87% faster than their non-staged counterparts [McGrady 2015]. However, traditional virtual staging methods require expert 3D artist labor to first manually recreate structural properties of spaces through photographs and then stage curated products in the scene. Recent improvements in 3D capture have greatly reduced the technological barrier, enabling consumers to scan their homes with only a smartphone [Matterport 2021]. In this work, we present Home Studio, a mixed reality design tool for home staging. Our tool removes reliance on expert-level 3D knowledge to produce photo-realistic visualizations of the home and increase sense of presence for prospective buyers with an accompanying virtual reality experience.

## 2 RELATED WORKS

Different approaches to create photo-realistic composites of 3D models and photographs have been explored. Most photo-based applications focus on accurate reconstruction of scene layouts [Adobe 2021; Fournier et al. 1993; Karsch et al. 2014] or visual coherency of lighting [Gardner et al. 2019; Kán and Kafumann 2019], but have limited interior design capabilities because the output is limited to a single image. Some mixed reality applications [Park et al. 2016; Yue et al. 2017] utilize an in-situ staging approach to provide contextualization without scene reconstruction, letting customers see products in their space using augmented reality on their phones. However, mobile hardware constraints prevent photorealism [Ramos et al. 2014]. Online design services [Modsy 2021; Rooomy 2021] offer photo-realistic renders of staged spaces, but are reliant on expert 3D knowledge, leading to turnaround times in the order of days. In this work, we reduce the level of expertise required to design

photo-realistic room-scale scenes by extending prior art in the proxy distributed rendering space [Pomi 2005], while still providing for the immersion of mixed reality counterparts at the scale of a full property.

### 3 HOME STUDIO



**Figure 2:** Users can browse Wayfair’s comprehensive catalogue and stage furniture throughout the Matterport model, enhancing a virtual tool or helping them plan future purchases.



**Figure 3:** Nine examples of photo-realistic composites created by staging public properties on Redfin with the Home Studio tool.

Home Studio is a web based tool implemented using the Matterport SDK. The tool acts as a plugin to any Matterport scene viewer, enabling users to drag and drop furniture from Wayfair’s catalogue onto the scene. Our plugin makes any Matterport scene actionable. This includes properties previously scanned and available publicly in sites such as Redfin or lower resolution scans that can be captured through the Matterport capture application with an iPhone. Home Studio seamlessly integrates with the base Matterport tour experience which supports user movement to different 360 shots captured throughout a property. Our tool lets customers (1) browse a comprehensive catalogue of products by category, (2) select a product(s) to drop onto the scene, (3) drag the model to displace it around surfaces in the scene (including dropping products on walls

or on top of other products), (4) rotate the product to change its orientation, and (5) capture a specific view of the scene and render it.

Home Studio creates photo-realistic renders of the user’s viewpoint within the Matterport scene by reconstructing the view in a remote rendering service. When the user requests a render, our tool registers basic topology of the view which includes: the transform of surface planes (i.e. floors & walls), the camera’s transform, the camera’s field of view, and the transform of objects within the camera’s frustum. These parameters are used to generate a 3DS Max scene which replaces low-res models of furniture used in the web experience with high resolution analogues. The HDRI 360 image of the current waypoint is used as the environment map of the reconstructed scene to apply image based lighting to the final render. The reconstructed scene is then rendered using V-Ray; progressive frames are fed back to the client using Socket.io.

Spaces staged using Home Studio can also be experienced in virtual reality. Users can connect a headset such as a Google Cardboard or Oculus Quest to dive into the scene. While immersed in the space, the user can look around akin to a 360 viewer and move between different waypoints captured in the 3D space. Additionally, users can save and load in existing projects to continue iterating on previous designs.

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